

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>

To do this, see how many minutes a regular GB takes to download, calculate how much of the download is lost with the reinstall, and use total lost download time + time to get to 40% and the time for the reinstall to determine time it takes total.

1 GB time = 1 / 2 minute

200 GB time = 200 * 1 / 2 = 100 minutes to download.

40% download time = 40% * 100 = 40 minutes.

200 GB * 0.6 = 120 GB for 60% of download.

120 GB takes 120 * 0.5 = 60 minutes.

Total download time = 20 min reinstall + 60 min to get to 60% + 20 min from 60% to 100% = 60 min

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</think>

<answer>100</answer>

